Squares, Cubes and Roots

Concept

Squares and cubes describe repeated multiplication. Roots undo those powers.

Perfect square

$n^2 = n \times n$; perfect cube: $n^3 = n \times n \times n$.

Principal root

$\sqrt{a}$ is non-negative. $\sqrt[3]{a}$ can be negative.

Estimate

Place a number between two nearby perfect squares.

Watch the sign

$\sqrt{49} = 7$, but $x^2 = 49$ has solutions ± 7.

Example 1

Find the side and perimeter of a square with area $196\ m^2$.

Solution

1. Side $= \sqrt{196} = 14$ m.

2. Perimeter $= 4 \times 14 = 56$ m.

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Example 2

Is 180 a perfect square? Estimate its square root.

Solution

1. $180 = 2^2 \times 3^2 \times 5$. The factor 5 is unpaired: not a perfect square.

2. $169 < 180 < 196$, so $13 < \sqrt{180} < 14$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find $\sqrt{225}$, $\sqrt[3]{512}$ and $\sqrt[3]{(-125)}$.

2. Find $\sqrt{(25/144)}$.

3. Between which whole numbers is $\sqrt{85}$?

4. A square has perimeter 52 cm. Find its area.

5. Is 432 a perfect cube? Explain using prime factors.

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Practice answers | check after trying

1. 15, 8 and -5; squaring or cubing returns the original values.

2. 5/12.

3. Between 9 and 10 because $81 < 85 < 100$.

4. Side = 52/4 = 13 cm; area = $13^2 = 169$ cm^2.

5. No. $432 = 2^4 \times 3^3$, so the four factors of 2 cannot all form groups of three.